Two-step Lyapunov decrease at a logarithmic boundary for a nonconvex Douglas–Rachford iteration

Abstract

We consider the Douglas–Rachford relation for a union of two affine lines and a third line, retaining every nearest-line projection at a tie. At the nonperpendicular logarithmic equality boundary of the product-energy criterion of Giladi and Rüffer, their product energy is nonincreasing but has nonfeasible equality transitions. We determine all such transitions: each affine branch has one contact on a projection bisector, and no equality successor is another equality input. Consequently the same product energy decreases strictly along every two-step path starting outside the feasible pair. Its properness and the closed graph of the full relation imply boundedness, decay of the energy to zero, and approach to the feasible pair for every trajectory. The argument uses exact contact geometry and a compact three-point limit passage.

Keywords: Douglas–Rachford iteration; product Lyapunov function; projection ties; two-step decrease

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1 The relation and the boundary result

Let $p_1 = (-1/2, 0)$, $p_2 = (1/2, 0)$ and $B = \mathbb{R} \times \{0\}$. For $i = 1, 2$, let A_i be the complete affine line through p_i with direction $(\cos\theta_i, \sin\theta_i)$, and put $A = A_1 \cup A_2$. Throughout, assume

$$0 < \theta_1 < \frac{\pi}{2}, \qquad \theta_1 < \theta_2 < \pi, \qquad \theta_2 \neq \frac{\pi}{2}. \tag{1}$$

Write $s_i = \sin\theta_i$, $c_i = \cos\theta_i$, $C = (1 + s_1)(1 + s_2)$, and impose the exact boundary condition

$$(\log C)^2 = \log(c_1^2)\log(c_2^2). \tag{2}$$

Then $C > 1$ and $0 < c_i^2 < 1$. In particular,

$$\alpha = -\frac{\log C}{\log(c_1^2)} > 0, \qquad V(x) = \|x - p_1\|^{2\alpha}\|x - p_2\|^2 \tag{3}$$

is well defined. Set $P = \{p_1, p_2\} = A \cap B$.

The clockwise affine branches and their admissibility conditions are

$$T_i(x) = p_i + c_i R_i(x - p_i), \qquad R_i = \begin{pmatrix} c_i & s_i \\ -s_i & c_i \end{pmatrix}, \tag{4}$$

$$\ell_1(x) = s_1(x_1 + 1/2) - c_1 x_2, \qquad \ell_2(x) = s_2(x_1 - 1/2) - c_2 x_2,$$

$$T(x) = \{T_i(x) : |\ell_i(x)| \le |\ell_{3-i}(x)|,\ i = 1, 2\}. \tag{5}$$

Here $|\ell_i(x)| = \operatorname{dist}(x, A_i)$. Writing $R_D = 2P_D - I$, where P_D is the full nearest-point projection onto D, (5) is precisely the Douglas–Rachford relation $\frac{1}{2}(I + R_B R_A)$, with A projected first. Both outputs are retained at every tie; identical outputs represent one element of the value set. A trajectory is any sequence $x_{k+1} \in T(x_k)$, with arbitrary $x_0 \in \mathbb{R}^2$ and arbitrary subsequent admissible choices.

Theorem 1. *Under* (1)–(2), *the energy* (3) *satisfies* $V(y) \le V(x)$ *for every* $y \in T(x)$. *Moreover,*

$$V(z) < V(x) \quad \textit{whenever } x \notin P, \quad y \in T(x), \quad z \in T(y). \tag{6}$$

Every initial point admits an infinite trajectory. Every trajectory is bounded and satisfies

$$V(x_k) \longrightarrow 0, \qquad \operatorname{dist}(x_k, P) \longrightarrow 0. \tag{7}$$

Its omega-limit set is nonempty, compact and contained in P. *The only trajectories with constant energy at every step are* $x_k = p_1$ *and* $x_k = p_2$.

The map, product construction and cross-energy circles are due to Giladi and Rüffer [1]. Their Theorem 5.3, equations (18)–(24), proves one-step contraction in the strict coefficient region

$$(\log C)^2 < \log(c_1^2)\log(c_2^2),$$

and handles perpendicular cases separately. At equality, both coefficient bounds become one. Here we classify all legal equality transitions and derive strict two-step loss for the unchanged product V.

All source numbers refer to arXiv:1708.08697v2, dated 16 October 2018 [1]. The final journal main body [2] has not been inspected or used for this comparison. The symmetric minimum-energy construction and the reverse-order result in [1] are prior results. We assert feasible-set capture only, with no selected-point convergence, rate, robustness, or extension beyond the stated boundary.

2 The legal cross-energy contacts

Let $e_2 = (0, 1)$ and introduce

$$m = \frac{\theta_1 + \theta_2}{2}, \qquad \delta = \frac{\theta_2 - \theta_1}{2}, \qquad n = (\sin m, -\cos m), \qquad t = (\cos m, \sin m).$$

Thus n, t are orthogonal unit vectors, $\sin m > 0$ and $0 < \delta < \pi/2$. Put

$$b = \tfrac{1}{2}\cos m \tan\delta, \quad a = \tfrac{1}{2}\sin m \cot\delta, \quad H_2 = \{x : \langle n, x\rangle = b\}, \quad H_1 = \{x : \langle t, x\rangle = a\}. \tag{8}$$

The full tie set is $H_1 \cup H_2$, since

$$\ell_1 + \ell_2 = 2\cos\delta(\langle n, x\rangle - b), \qquad \ell_1 - \ell_2 = -2\sin\delta(\langle t, x\rangle - a). \tag{9}$$

Lemma 1. *For each legal branch,*

$$\|T_i(x) - p_{3-i}\|^2 \le C\|x - p_{3-i}\|^2. \tag{10}$$

Equality holds for a legal T_1 branch exactly at $q_1 = z_2 - r_2 n$, and for a legal T_2 branch exactly at $q_2 = z_1 + r_1 n$, where

$$\begin{aligned} z_1 &= p_1 + \frac{c_2 s_2}{C - c_2^2} e_2, & r_1 &= \frac{\sqrt{C}\, s_2}{C - c_2^2}, \\ z_2 &= p_2 - \frac{c_1 s_1}{C - c_1^2} e_2, & r_2 &= \frac{\sqrt{C}\, s_1}{C - c_1^2}. \end{aligned} \tag{11}$$

Both q_i lie on H_2 and outside P.

Proof. The denominators and radii in (11) are positive, because

$$C - c_i^2 = (1 + s_i)(s_1 + s_2) > 0. \tag{12}$$

Completing the square gives

$$\begin{aligned} \|T_2(x) - p_1\|^2 - C\|x - p_1\|^2 &= (C - c_2^2)(r_1^2 - \|x - z_1\|^2), \\ \|T_1(x) - p_2\|^2 - C\|x - p_2\|^2 &= (C - c_1^2)(r_2^2 - \|x - z_2\|^2). \end{aligned} \tag{13}$$

These are the circles of [1, Proposition 5.2, equations (14)–(15)] with parameter $\rho = C$. We record their precise contacts with the legal regions rather than infer equality cases from an inequality.

Elementary trigonometric identities give

$$\sqrt{C} = \cos\delta + \sin m, \qquad C + c_1 c_2 = 2\cos\delta\sqrt{C}. \tag{14}$$

Indeed $\cos\delta + \sin m = (1+u)(1+v)/\sqrt{D}$ for $u = \tan(\theta_1/2)$, $v = \tan(\theta_2/2)$ and $D = (1+u^2)(1+v^2)$, which also verifies the positive square root. Substitution in (11) yields

$$\langle n, z_1\rangle - b = -r_1, \qquad \langle n, z_2\rangle - b = r_2, \tag{15}$$

$$\operatorname{dist}(z_1, H_1)^2 - r_1^2 = \frac{C s_1 s_2^3}{\sin^2\delta\,(C - c_2^2)^2} > 0, \tag{16}$$

$$\operatorname{dist}(z_2, H_1)^2 - r_2^2 = \frac{C s_1^3 s_2}{\sin^2\delta\,(C - c_1^2)^2} > 0. \tag{17}$$

For completeness, these substitutions use $\sin m = (u+v)/\sqrt{D}$, $\cos m = (1-uv)/\sqrt{D}$, $\sin\delta = (v-u)/\sqrt{D}$, $\cos\delta = (1+uv)/\sqrt{D}$, together with $\sin\theta = 2w/(1+w^2)$ and $\cos\theta = (1-w^2)/(1+w^2)$ for $w = \tan(\theta/2)$. Thus (15)–(17) are rational identities after positive denominators are cleared.

The center z_1 is strictly closer to A_1 than to A_2, since

$$\operatorname{dist}(z_1, A_1) = \frac{|c_1 c_2| s_2}{C - c_2^2} < \frac{C s_2}{C - c_2^2} = \operatorname{dist}(z_1, A_2).$$

Similarly, z_2 is strictly closer to A_2:

$$\operatorname{dist}(z_2, A_2) = \frac{|c_1 c_2| s_1}{C - c_1^2} < \frac{C s_1}{C - c_1^2} = \operatorname{dist}(z_2, A_1).$$

By (15)–(17), each open ball $B(z_i, r_i)$ avoids both tie lines, and its closed ball meets their union at its unique H_2 tangency. The ball is connected, so the sign of $\ell_1^2 - \ell_2^2$ throughout its interior is the sign at its center. Every boundary point other than the tangency has the same strict sign, by continuity and its exclusion from the tie set.

Consequently $B(z_2, r_2)$ lies in the region where T_1 is illegal, and its boundary meets the legal T_1 region only at $z_2 - r_2 n$. The corresponding assertion for T_2 gives $z_1 + r_1 n$. Together with (13), this proves the bound and its full equality classification. Both contacts are ties. At p_1 the line distances are $0, s_2$, and at p_2 they are $s_1, 0$; hence neither feasible point is a tie. ∎

3 All zero drops and their successors

The half-angle parameters satisfy $0 < u < 1$, $v > u$ and $v \neq 1$. The contacts in Lemma 1 take the explicit form

$$q_1 = \left(\frac{(u-1)(uv-1)}{2(u+1)(uv+1)}, -\frac{u(v-1)}{(u+1)(uv+1)} \right), \tag{18}$$

$$q_2 = \left(-\frac{(v-1)(uv-1)}{2(v+1)(uv+1)}, \frac{v(u-1)}{(v+1)(uv+1)} \right). \tag{19}$$

In particular,

$$q_1 - q_2 = \frac{\big((uv-1)^2, -(u+v)(uv-1)\big)}{(u+1)(v+1)(uv+1)}. \tag{20}$$

Thus $q_1 = q_2$ exactly when $uv = 1$. Both second coordinates in (18)–(19) are nonzero in the stated domain.

Proposition 3.1. The complete legal equality relation is

$$\{(x, y) : y \in T(x),\ V(y) = V(x)\} = \{(p_1, p_1), (p_2, p_2), (q_1, T_1(q_1)), (q_2, T_2(q_2))\}. \tag{21}$$

For $i = 1, 2$, the equality successor $T_i(q_i)$ lies outside $P \cup \{q_1, q_2\}$. Therefore every non-feasible equality transition is followed by strict loss on every legal next branch.

Proof. Equation (4) gives the exact local contraction

$$\|T_i(x) - p_i\|^2 = c_i^2 \|x - p_i\|^2. \tag{22}$$

The boundary condition and the definition of α give

$$(c_1^2)^\alpha C = 1, \qquad c_2^2 C^\alpha = 1. \tag{23}$$

Combining (22) with Lemma 1 therefore proves $V(T_i(x)) \le V(x)$ for every legal branch. For $x \notin P$, both factors in $V(x)$ are positive. Equality is equivalent to equality in (10); the positive exponent α preserves this equivalence for branch T_2 as well. The only such inputs are q_1 and q_2 on their respective branches. At p_i, only branch T_i is legal and $T_i(p_i) = p_i$. This proves (21), including all projection ties.

To exclude consecutive nonfeasible equality steps, observe first that

$$\langle T_i(x) - p_i, T_i(x) - x \rangle = 0. \tag{24}$$

Indeed, for $w = x - p_i$, one has $\|c_i R_i w\|^2 = c_i^2 \|w\|^2$ and $c_i \langle R_i w, w \rangle = c_i^2 \|w\|^2$. If $uv \neq 1$, substitution of (18)–(19) gives

$$\langle q_2 - p_1, q_2 - q_1 \rangle = -\frac{v(1+u^2)(v-1)(uv-1)}{(u+1)(v+1)^2(uv+1)^2} \neq 0, \tag{25}$$

$$\langle q_1 - p_2, q_1 - q_2 \rangle = -\frac{u(u-1)(1+v^2)(uv-1)}{(u+1)^2(v+1)(uv+1)^2} \neq 0. \tag{26}$$

All denominators are positive, and all numerator factors are nonzero under the present assumptions. Equation (24) now excludes $T_1(q_1) = q_2$ and $T_2(q_2) = q_1$. Also,

$$\det(I - c_i R_i) = s_i^2 > 0,$$

so p_i is the only fixed point of T_i. Thus $T_i(q_i) \neq q_i$. If $uv = 1$, this same fixed-point argument excludes either successor from the common contact $q_1 = q_2$. Finally, $V(T_i(q_i)) = V(q_i) > 0$ excludes a successor in P. By (21), every legal step from such a successor strictly decreases V. ∎

If the contacts are distinct, the other legal branch at each q_i already strictly decreases V. If they coincide, both legal branches preserve V at the common contact, but both are followed by strict loss. Thus a one-step strictness assertion would fail, even though every two-step path has the strict decrease required in Theorem 1.

4 Compact trajectories and capture

Proof of Theorem 1. There is always a minimum among $|\ell_1(x)|$ and $|\ell_2(x)|$, so $T(x)$ is nonempty and contains at most two points. Successive choices therefore define a trajectory for every nonnegative integer index. The relation has a closed graph: if $x_j \to x$, $y_j \to y$ and $y_j \in T(x_j)$, pass to an infinite subsequence on which one branch index i is fixed. Continuity gives $y = T_i(x)$ and $|\ell_i(x)| \le |\ell_{3-i}(x)|$, hence $y \in T(x)$. This uses the weak legal inequalities and remains valid at ties. The two affine branches also map each compact set into a bounded set.

The energy is continuous, nonnegative, and vanishes exactly on P. Furthermore,

$$V(x) \ge \operatorname{dist}(x, P)^{2\alpha+2}. \tag{27}$$

Thus $V(x) \to +\infty$ as $\|x\| \to \infty$, and all its sublevel sets are compact. No differentiability of V at P is needed.

Nonincrease was proved in Proposition 3.1. Given $x \notin P$, $y \in T(x)$ and $z \in T(y)$, if $V(y) < V(x)$, then $V(z) \leq V(y) < V(x)$. Otherwise the successor exclusion in that proposition gives $V(z) < V(y) = V(x)$. This proves (6). At each p_i, the unique legal successor is p_i itself. Hence the only trajectories with constant energy at every step are the two feasible constant trajectories.

Fix any trajectory. The nonincreasing sequence $V(x_k)$ has a limit $\lambda \geq 0$, and the entire trajectory lies in the compact set

$$K = \{x \in \mathbb{R}^2 : V(x) \leq V(x_0)\}.$$

Its omega-limit set is consequently nonempty and compact. Let x be any accumulation point, with $x_{k_j} \to x$ and $k_j \to \infty$. By compactness of $K \times K$, a further subsequence satisfies

$$x_{k_j+1} \longrightarrow y, \qquad x_{k_j+2} \longrightarrow z.$$

Applying closedness of the one-step graph twice gives $y \in T(x)$ and $z \in T(y)$. Continuity of V, and convergence of the same scalar energy sequence along each of these subsequences, imply

$$V(x) = V(y) = V(z) = \lambda.$$

Strict two-step decrease forces $x \in P$. Thus the omega-limit set is contained in P, and its nonemptiness gives $\lambda = 0$. Equation (27) now yields $\operatorname{dist}(x_k, P) \to 0$, completing (7). ∎

The accompanying symbolic checker verifies the rational contact identities. The domain signs, legal-region completeness and compactness arguments are proved above.

References

[1] O. Giladi and B. S. Rüffer, A Lyapunov function construction for a non-convex Douglas–Rachford iteration, arXiv:1708.08697v2, 16 October 2018. `https://arxiv.org/abs/1708.08697v2`. Version used for all source comparisons.

[2] O. Giladi and B. S. Rüffer, A Lyapunov Function Construction for a Non-Convex Douglas–Rachford Iteration, *Journal of Optimization Theory and Applications* **180** (2019), 729–750. `https://doi.org/10.1007/s10957-018-1405-3`. Journal publication; comparisons use [1].